

Summary and Critical Analysis of an Article - TP547

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Abstract—This manuscript summarizes and critically analyzes an article on automatic digital modulation classification using constellation density and deep learning. The analyzed work converts received-signal constellations into color images, applies geometry-based masking filters, and uses deep neural networks to classify amplitude, phase, and quadrature modulation schemes. The article is relevant to digital modulation and baseband communication because it links complex-symbol modeling, constellation diagrams, additive noise, signal-to-noise ratio, and automatic decision-making. However, its contribution is partly incremental, its validation is limited to synthetic conditions, its mathematical formulation is insufficient for full reproducibility, and its comparisons do not fully resolve concerns about experimental fairness. Thus, the method is promising as a proof of concept, but broader channel models, deeper ablation studies, statistical confidence analysis, complexity evaluation, and open implementation details are required before robustness can be claimed in practical communication systems.

Index Terms—digital modulation, baseband communication, constellation density, deep learning, critical analysis.

I. INTRODUCTION

Identifying the modulation format of an unknown received signal is a fundamental objective in automatic modulation classification (AMC), as it enables the transition from signal interception to reliable demodulation in non-cooperative communication scenarios [1]. This task is central to digital communications because the modulation scheme determines how bits are represented as symbols, directly affecting spectral efficiency, receiver complexity, and robustness to noise. The selected article is therefore mainly related to digital modulation, while also addressing baseband communication through its use of complex received symbols and constellation diagrams. Additionally, it has an indirect connection with quantization, since the complex plane is discretized into a finite grid before classification.

The Institute of Electrical and Electronics Engineers (IEEE) Author Center [2] recommends that technical articles contain a specific title, self-contained abstract, clear introduction, reproducible methodology, well-supported results, and conclusions that explain contributions and limitations. It also indicates that reviewers evaluate scope, validity, data quality, clarity, compliance, advancement, and novelty. These criteria guide this work, which summarizes the selected paper and evaluates its writing, contribution, equations, and experimental demonstration.

Kumar et al. [3] proposed *Automatic Modulation Classification Based on Constellation Density Using Deep Learning*,

published in IEEE Communications Letters in 2020. The paper develops a classifier based on constellation-density images, masking filters, and deep neural networks. The following sections analyze the work from technical and editorial perspectives, emphasizing methodological clarity, scientific contribution, experimental strength, and limitations relevant to peer review.

II. SELECTED ARTICLE AND THEMATIC SCOPE

The selected article is a short journal letter on AMC using deep learning (DL). It belongs primarily to digital modulation because it classifies amplitude-shift keying (ASK), phase-shift keying (PSK), and quadrature amplitude modulation (QAM) signals. It also relates to baseband communication because its model uses complex received symbols represented in the constellation plane. Bandpass communication appears only as context, since the introduction mentions unknown radio frequency (RF) signals, while the modeling and experiments use the equivalent baseband domain.

The article does not address sampling and quantization in the classical sense of analog-to-digital conversion, sampling rate, aliasing, or amplitude quantization. Nevertheless, its preprocessing performs spatial quantization by normalizing constellation points, restricting them to a square region, and assigning them to a 100×100 grid. Therefore, the article is strongly linked to digital modulation and uses a discrete constellation representation as input to machine-learning models.

III. SUMMARY OF THE ARTICLE

A. Problem, Motivation, and Declared Contribution

The analyzed article addresses automatic identification of the modulation format of a received signal without prior explicit knowledge of the transmitted scheme. It presents likelihood-based classifiers as accurate but computationally demanding when channel parameters are unknown. It then presents feature-based methods as simpler but dependent on manual feature selection. This limitation motivates DL, since deep networks can learn discriminative patterns from data.

The declared contribution has three components. First, the paper introduces the constellation density matrix (CDM), which captures the local spatial distribution of constellation points. Second, it designs masking filters that suppress regions considered noisy and convert the CDM into three-channel color images. Third, it uses a hierarchical classifier with residual network (ResNet)-50 and Inception ResNet V2. Thus, the

work converts a digital-communication classification problem into an image-recognition problem.

B. Signal Model and Basic Equations

The article considers ASK, PSK, and QAM signals transmitted through an additive white Gaussian noise (AWGN) channel. The received signal is modeled as

$$y[n] = x[n] + \eta[n], \quad n = 1, 2, \dots, N, \quad (1)$$

where $y[n]$ is the received complex symbol, $x[n]$ is the transmitted complex symbol, and $\eta[n]$ is zero-mean additive white Gaussian noise with variance σ_η^2 . For unit-power constellations, the signal-to-noise ratio (SNR) is defined as

$$\text{SNR} = 10 \log_{10} \left(\frac{1}{\sigma_\eta^2} \right). \quad (2)$$

These equations are suitable for an initial proof of concept because they isolate additive noise. However, they describe only an idealized channel. The paper does not model fading, frequency offset, phase noise, timing error, pulse shaping, amplifier nonlinearities, or synchronization impairments. Therefore, the reported performance should be interpreted mainly for a controlled synthetic baseband scenario.

C. Image Formation Using the Constellation Density Matrix

The article converts each received constellation into a $3 \times 100 \times 100$ color image. After normalization by average energy, only a 4×4 region of the complex plane is considered and divided into 100×100 equal grid sections. The number of points in each section is counted and scaled to $[0, 255]$, forming the CDM. Each CDM is then filtered by Boolean masks, and each mask output becomes one red green and blue (RGB) channel:

$$\text{CDM}(i, j) \cdot M_k^m = \begin{cases} \text{CDM}(i, j), & M_k^m = 1, \\ 0, & M_k^m = 0, \end{cases} \quad (3)$$

where M_k^m is the m -th mask of filter F_k .

Table I summarizes the filters reported in the article. It shows that the method does not rely only on automatic feature learning, since it embeds prior geometric knowledge about expected constellation locations. Therefore, the approach is better described as a hybrid method combining handcrafted domain knowledge with DL.

D. Hierarchical Classification and Training

He et al. [4] introduced residual networks to address optimization difficulties in deep models through shortcut connections. Szegedy et al. [5] developed Inception-ResNet architectures combining inception modules and residual connections. These works justify the use of ResNet-50 and Inception ResNet V2 as image-classification backbones, as done by the analyzed article.

The classifier has three stages. The first separates signals into broad ASK, PSK, and QAM groups. The second distinguishes modulation orders within each group. The third uses two binary classifiers to refine higher-order QAM decisions.

This hierarchy simplifies the eight-class problem, but it also creates error propagation because an early wrong decision may prevent later recovery of the correct class.

The training dataset covers SNR values from -4 dB to 30 dB with a 1 dB step, and from 35 dB to 70 dB with a 5 dB step. For each modulation, fifty images are generated per SNR, resulting in 2150 images per class. The implementation uses Python, Keras, and an NVIDIA DGX-2 graphics processing unit (GPU). The article reports learning rate 0.0001, minibatch size 64, and dropout probability 0.5.

E. Reported Results

The article reports that ResNet-50 and Inception ResNet V2 achieve reliable classification above approximately 5 dB for most modulations. The main exception is the confusion between 16-QAM and 64-QAM in the final stage. The paper also compares the proposed method with genetic programming with k-nearest neighbors (GP-KNN), maximum-likelihood classification, and a two-dimensional Kolmogorov–Smirnov (K–S) classifier under selected conditions.

Aslam et al. [6] proposed a classifier based on genetic programming and k-nearest neighbors (KNN), which serves as a classical baseline. Chen et al. [7] studied maximum-likelihood modulation classification in non-Gaussian channels, providing another comparison point. Peng et al. [8] had already used constellation diagrams and DL, making it necessary to verify whether the novelty of the analyzed article truly lies in the CDM, masking filters, and hierarchical decision strategy.

IV. CRITICAL ANALYSIS

This section evaluates the article according to the assignment requirements and typical peer-review criteria. It considers writing quality, contribution relevance, equation clarity, and result strength. Overall, the article is technically interesting and relevant to digital modulation, but its short format leaves important gaps in reproducibility, generalization, and experimental validation.

A. Strengths

The first strength is the relevance of the problem. AMC is important in cognitive radio, spectrum monitoring, non-cooperative receivers, and adaptive communication systems. Since the article addresses ASK, PSK, and QAM, it is appropriate for a course assignment on digital modulation and baseband communication.

The second strength is the integration of domain knowledge with DL. Although the paper uses deep networks, the masks are based on constellation geometry. This can reduce the influence of noisy regions and emphasize spatial patterns meaningful in communication theory. Thus, the approach is not a blind convolutional neural network (CNN) application, but a combination of signal-processing intuition and image-based classification.

The third strength is the visual organization of the methodology. The main figure shows the classification hierarchy, filters, generated images, and deep-learning blocks. Although dense,

TABLE I
SUMMARY OF THE MASKING FILTERS USED TO CONVERT THE CDM INTO RGB IMAGES, ADAPTED FROM THE ANALYZED ARTICLE.

Filter	Red channel	Green channel	Blue channel
F_1	Rectangle: 11×100	Ring: $R_1 = 27, R_2 = 37$	Four rectangles: 9×100
F_2	Square: 100×100	Two circles: $R = 8$	Four circles: $R = 6$
F_3	Square: 100×100	Four circles: $R = 18$	Eight circles: $R = 10$
F_4	Square: 100×100	Square: 100×100	Square: 100×100
F_5	Square: 100×100	Eight circles: $R = 10$	Sixteen circles: $R = 6$
F_6	Square: 100×100	Thirty-two circles: $R = 8$	Sixty-four circles: $R = 4$

it helps explain the complete pipeline. The comparison with and without masks is also valuable because it provides an initial ablation study.

The fourth strength is the comparison with previous classifiers. The article does not present only isolated accuracy curves, but relates the method to existing approaches. This strengthens the contribution narrative because absolute accuracy is difficult to interpret without baselines. The use of two deep architectures also suggests that the preprocessing idea is not tied to one model.

The fifth strength is concision. As a letter, the article presents an introduction, signal model, preprocessing, training, results, and conclusion in a limited space. The structure is logical, and the central idea is understandable. Therefore, the article is partially well written, although several methodological details are omitted.

B. Weaknesses

1) *The Contribution Is Potentially Incremental:* The article states in its abstract, “Classification accuracy achieved demonstrates better performance compared to many existing classifiers in the literature.” This is vague because *many existing classifiers* is not clearly delimited by methods, assumptions, or parameters. Since previous work had already used constellation diagrams as images for DL-based modulation classification, the novelty is not the use of constellation images alone, but the combination of CDM, masks, and hierarchy.

A stronger paper would define the contribution more precisely as geometry-guided masking of constellation-density images. It should compare directly with prior constellation-image methods under the same dataset, modulation classes, number of symbols, and SNR values. It should also include separate ablations for the CDM, masks, hierarchy, and neural architecture to show the measurable gain of each component.

2) *The Channel Model Is Too Narrow:* The experiments use only an AWGN channel, normalized symbols, and artificial constellations. This controlled setup is useful, but far from practical receivers, which may suffer phase rotation, frequency offset, timing error, multipath fading, co-channel interference, pulse-shaping effects, and hardware impairments. Since the masks assume specific geometric regions, rotations or scaling errors may move points away from the expected positions.

The paper could be improved by adding Rayleigh, Rice, and frequency-selective multipath channels. It should also test carrier-frequency offset, phase noise, gain mismatch, and in-phase/quadrature imbalance. Stronger validation would in-

clude software-defined-radio measurements, while augmentation with random rotations, translations, scaling, and gain variations would reduce dependence on perfectly aligned synthetic constellations.

3) *The Equations Are Not Sufficient for Reproduction:* The signal model and SNR definition are clear, but the preprocessing mathematics is incomplete. Eq. (1) defines the received signal, Eq. (2) defines SNR, and Eq. (3) describes masking. However, the paper does not fully define grid boundaries, mapping from complex points to cells, energy normalization, scaling to $[0, 255]$, treatment of points outside the 4×4 region, or exact mask geometry.

A better version would provide a formal CDM construction algorithm defining the complex-plane region, grid spacing, cell-indexing rule, and counting operation. It should specify whether normalization uses the maximum value, the number of symbols, or another factor. Finally, each mask should be expressed through geometric inequalities, making the method implementable without visual interpretation of the figure.

4) *Reproducibility Is Limited:* The article reports the programming language, library, GPU, learning rate, minibatch size, and part of the network structure. However, it omits code, dataset files, random seeds, number of epochs, optimizer, loss function, train-validation-test split, stopping criterion, frozen layers, and the exact signal-generation procedure. These omissions hinder independent reproduction.

The solution is to publish signal-generation, preprocessing, training, and evaluation code in an open repository. The paper should also report seeds, dataset partitions, optimizer settings, epochs, early stopping, and a complete hyperparameter table. Providing generated CDM images or complex symbol sequences would allow identical-condition comparisons.

5) *The Experimental Design May Overestimate Generalization:* The article states that fifty images per SNR are generated for training and that testing uses one thousand signal realizations. However, it is unclear whether training and testing share the same SNR grid, assumptions, and generation settings. If both sets come from the same synthetic process, the model may learn simulator artifacts rather than robust modulation properties.

A stronger design would include out-of-distribution tests. The model could be trained on selected SNR values and tested on unseen intermediate values. It could also be trained with one signal generator and tested with another using different mappings, pulse shapes, phase offsets, and symbol-window lengths. A blind test with real measured signals would provide

stronger evidence of practical generalization.

6) *The Comparisons Are Not Fully Fair:* The comparisons are useful but incomplete. Not all baselines are evaluated for all eight modulation classes, and some were designed for different channel assumptions, including Gaussian mixture model (GMM) conditions. There is also a distinction between results using very long signals and comparisons reported for 512 symbols. Thus, it is difficult to know whether the gain comes from the method, symbol count, modulation subset, or channel assumptions.

A fairer comparison would reimplement all baselines under identical conditions, using the same classes, number of symbols, SNR values, and train-test protocol. The article should also report complexity, inference time, and memory usage, since deep networks may be more expensive than statistical or feature-based classifiers.

7) *The Results Lack Statistical Analysis:* The article mainly reports classification accuracy. Although useful, accuracy is insufficient for evaluating a hierarchical modulation classifier. The paper does not provide confidence intervals, standard deviations, hypothesis tests, detailed confusion matrices, or per-stage error analysis. The confusion between 16-QAM and 64-QAM is mentioned but not deeply investigated.

The results would be stronger with confusion matrices at representative SNR values, as well as precision, recall, and F1-score per class. Repeated experiments with independent seeds would allow confidence interval estimation. The classifier should also be evaluated versus the number of received symbols, since practical receivers may require decisions from shorter observation windows.

8) *The Neural Architectures May Be Too Heavy:* ResNet-50 and Inception ResNet V2 were designed for complex natural-image recognition, while constellation-density images are synthetic and structured. The paper also adds eight fully connected layers to the pretrained networks. This may increase computational cost and overfitting risk, especially with a relatively modest number of training images.

The paper should compare these models with a lightweight CNN trained from scratch. It should also reduce the fully connected layers and evaluate the effect through ablation. Reporting trainable parameters, floating-point operations, training time, and inference latency would show whether the accuracy gain justifies the complexity. Smaller models may be preferable for real-time receivers.

9) *The Fixed Masks Are a Potential Fragility:* The masks are the most original part of the paper, but also a possible weakness. They assume constellations are centered, normalized, and aligned with expected geometry. In real systems, phase rotation, gain-estimation error, frequency offset, and channel distortion may deform the constellation. Fixed masks could then suppress useful information or emphasize wrong regions.

Possible solutions include learnable masks through attention mechanisms or trainable spatial filters initialized with the proposed masks. The receiver could also estimate and compensate for gain, phase, and frequency offset before forming the CDM.

Additional robustness could come from training with random rotations and scaling, or from using invariant descriptors such as radial distributions, angular histograms, and normalized geometric moments.

10) *The Hierarchical Classifier Propagates Errors:* The hierarchy simplifies classification, but introduces error propagation. If the first-stage classifier assigns the signal to the wrong family, later classifiers operate in the wrong branch, and the correct final class may become unreachable. Although the paper multiplies correct-classification probabilities across stages, it does not show detailed per-stage confusion matrices.

The paper should report the error distribution at each stage and compare the hierarchy with a direct eight-class classifier. A probabilistic hierarchy could evaluate the two most likely branches before the final decision, reducing irreversible early errors. Another option is a loss function that penalizes high-level family errors more strongly than within-family errors.

11) *Some Sentences Are Vague or Grammatically Weak:* The article states, “Reasons for accuracy improvement is discarding the noisy constellation points by masking filters.” The idea is understandable, but the sentence is grammatically weak and makes an insufficiently justified causal claim. The gain may result not only from discarding noisy points, but also from color-channel encoding, dimensionality reduction, class-dependent masks, the hierarchical structure, or training differences. In addition, there is a grammatical mistake in the sentence: the verb should be plural. The correct form is “Reasons for accuracy improvement are discarding the noisy constellation points by masking filters.”

A stronger version would state only that masks are associated with accuracy improvement under the simulated conditions. The authors should avoid attributing the entire improvement to a single cause without adequate evidence. The text would also benefit from technical editing, clearer causal language, and a more balanced discussion of when masks may fail.

12) *The Figures Are Dense:* The main methodology figure combines the classification hierarchy, filter outputs, mask operation, and neural-network blocks in one illustration. This compactness is useful in a short letter, but reduces readability. The result figure also contains several small accuracy plots, making precise comparisons difficult.

The article would be clearer if the methodology were split into separate figures for the hierarchy and masks. The results section would benefit from a table with numerical accuracy at selected SNR values. Higher-resolution plots, larger legends, and supplementary confusion matrices would make the conclusions easier to verify.

13) *The Contribution Is Only Partially Demonstrated:* The answer to whether the authors demonstrate the contribution in the results is only partially positive. The comparison with and without masks supports the claim that masks improve average performance. However, the paper does not isolate each component, since it does not compare raw constellation images against CDM images, separate the hierarchy from the neural architecture, or test sensitivity to mask dimensions.

TABLE II
DIRECT EVALUATION OF THE ARTICLE ACCORDING TO THE ASSIGNMENT CRITERIA.

Criterion	Judgment	Justification
Writing quality	Partially satisfactory	The structure is logical and concise, but some statements are vague, some sentences are grammatically weak, and limitations are not discussed deeply.
Contribution relevance	Relevant but incremental	The masking strategy is interesting, but constellation-image classification with DL had already been explored. The novelty requires stronger ablation studies.
Equations	Partially explained	The AWGN model and SNR definition are clear, but the CDM, spatial quantization, normalization, and mask geometry are not fully formalized.
Results	Partially convincing	The accuracy curves are promising, and the with-mask comparison is useful. However, the paper lacks confidence intervals, confusion matrices, realistic channels, and uniform baselines.
Contribution demonstration	Incomplete	The article shows that masks improve average performance, but it does not fully separate preprocessing, hierarchy, architecture, and simulation-condition effects.

A more convincing study would include a complete ablation table comparing raw constellation images, CDM without masks, CDM with masks, a single eight-class classifier, and the proposed hierarchy. The authors should also vary the grid size, mask radius, mask width, and number of symbols. These experiments would show whether the method is robust or tuned to one simulation setting.

C. Direct Evaluation of the Assignment Criteria

Table II summarizes the evaluation. Overall, the article is relevant and interesting, but it requires stronger evidence and clearer reporting to satisfy a rigorous reproducibility-oriented review.

V. CONCLUSION

The analyzed article is appropriate for a study on digital modulation because it classifies ASK, PSK, and QAM signals using baseband constellation information. Its main contribution is the combination of CDM image formation, geometry-based masking filters, and hierarchical DL classification. The method is clear as a proof of concept and shows promising results under AWGN conditions.

However, the contribution is partly incremental, the channel model is narrow, the equations are insufficient for full reproduction, and the comparisons could be more rigorous. The article would be strengthened by open code, formal CDM definitions, realistic channel models, statistical analysis, complexity evaluation, confusion matrices, and complete ablation studies. Therefore, the work is relevant and aligned with the course topic, but it does not yet prove robustness for practical digital communication systems.

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